

FuseLens: Long-Context Genomic Foundation Models for Robust DNA Breakpoint Detection

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December 15, 2025

Abstract

Gene fusions are critical oncogenic drivers, yet identifying their precise genomic breakpoints remains computationally challenging due to repetitive regions and complex rearrangements. Traditional alignment-based methods often fail to capture long-range dependencies in DNA sequences, leading to high false-positive rates. We introduce FuseLens, a novel deep learning framework leveraging HyenaDNA, one of the first applications of sub-quadratic ($O(N \log N)$) genomic foundation models to this task. This enables robust feature extraction across unprecedented 32kb genomic windows at single-nucleotide resolution. We propose a specialized Attention Pooling mechanism to localize breakpoint signals and introduce a formally defined synthetic negative augmentation framework. Furthermore, we formalize a domain adaptation protocol to integrate real RNA-Seq data with synthetic baselines. Benchmarking on a large-scale curated dataset of 195,143 sequences demonstrates strong performance with an AUC of 0.9904, alongside a Confidence-Gated Attention mechanism that localizes breakpoints without dense supervision, a critical step towards clinical reporting workflows.

1 Introduction

The detection of gene fusions—hybrid genes formed by chromosomal rearrangements—is a cornerstone of precision oncology. While tools like STAR-Fusion [5] and Arriba [6] have standardized detection from RNA-Seq data, they fundamentally rely on short-read alignment. This “alignment-first” paradigm struggles with ambiguous mappings in repetitive regions (e.g., *Alu* repeats) and often fails to capture the broader genomic context (intronic sequences, regulatory motifs) that characterizes true breakpoints.

Deep learning offers an alternative “sequence-first” paradigm. Early attempts utilized Convolutional Neural Networks (CNNs), such as FusionAI [2], but these architectures are limited by restricted receptive fields. Similarly, while Transformer-based models like BERT have revolutionized natural language processing, they are often ill-suited for high-resolution genomics due to quadratic computational complexity ($O(N^2)$). While recent sparse and linear-attention variants mitigate this cost, they remain impractical at single-nucleotide resolution over tens of kilobases. Gene fusions are often driven by distal genomic interactions; attempting to process such lengths with standard architectures is computationally intractable. Consequently, a new class of architecture is required—one capable of global context processing without the prohibitive cost of self-attention.

We present **FuseLens**, which adapts HyenaDNA [3], a sub-quadratic ($O(N \log N)$) genomic foundation model, for breakpoint detection. Unlike BERT, HyenaDNA utilizes implicit long convolutions to process **32kb** genomic windows at single-nucleotide resolution, enabling the capture of long-range structural dependencies that shorter-context models miss. FuseLens addresses the binary classification problem of determining whether a genomic window contains a true fusion breakpoint, and conditionally localizes the breakpoint position. By pairing this

backbone with a custom **Attention Pooling** head and a mathematically rigorous Synthetic Negative Augmentation strategy, FuseLens learns to distinguish true fusion breakpoints from biological noise with unprecedented accuracy over extended contexts.

2 Biological Background: Chimeric Transcripts in Cancer

Chimeric transcripts are RNA molecules formed by the joining of exons from two or more different genes. Understanding their diverse formation mechanisms is critical for distinguishing pathogenic drivers from benign artifacts [1].

2.1 Mechanisms of Formation

- **Read-Through (Cis-SAGE):** This mechanism occurs at the transcriptional level when RNA polymerase fails to recognize the stop signal of the upstream gene and continues transcribing into the neighboring gene’s coding region. The resulting fusion transcript is produced from an otherwise intact genomic sequence. Throughout this work, we use “read-through” and “Cis-SAGE” interchangeably to denote transcriptional chimeras without underlying DNA rearrangement.
- **Chromosomal Rearrangement:** This involves permanent DNA structural alterations, where DNA breaks and rejoins incorrectly. This category includes:
 - *Intra-chromosomal* inversions (e.g., *EML4-ALK*), where a segment on a single chromosome breaks, flips, and re-ligates, joining two genes.
 - *Inter-chromosomal* translocations (e.g., *BCR-ABL1*), where segments between two different chromosomes are swapped and joined, juxtaposing the two fusion partner genes.
- **Cis-Type Fusion (Structural Variant-Driven):** These fusions occur between genes located on the same chromosome (in *cis*), but are driven by a structural variant in the DNA, such as a large deletion or duplication, that physically removes the intervening sequence and brings the two genes into direct contiguity.

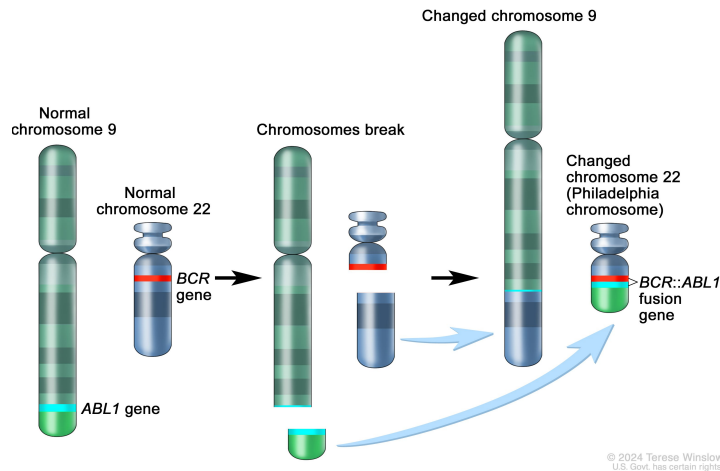


Figure 1: **The BCR-ABL1 Fusion Event.** Reciprocal translocation $t(9;22)$ creates a fusion gene on the derivative chromosome 22, characteristic of CML [14].

2.2 Examples for Key Oncogenic Fusions

We focus on detecting high-impact fusions characterized in the literature such as:

- **BCR-ABL1 (CML):** Resulting from the t(9;22) translocation (Philadelphia Chromosome). It creates a constitutively active tyrosine kinase driving Chronic Myeloid Leukemia [14].
- **TMPRSS2-ERG (Prostate Cancer):** A fusion of the androgen-responsive promoter of *TMPRSS2* with the oncogene *ERG*. Found in ~50% of prostate cancers [15].
- **SLC45A3-ELK4 (Prostate Cancer):** A prominent example of **Cis-Splicing**. This chimera can form via transcriptional read-through without any underlying DNA mutation [16].

2.3 Non-Cancerous Physiological Chimeras

Not all chimeric transcripts are pathogenic. Normal physiological chimeras occur in healthy cells and can drive evolutionary adaptation or protein diversity.

- **Benign Read-Throughs:** Transcripts like *ZFAND3-ZFAND6* occur in normal human tissues via Cis-SAGE (Splicing of Adjacent Genes) to create novel protein isoforms without genomic rearrangement [19].
- **Normal Trans-Splicing:** The *JAZF1-JJAZ1* RNA chimera, originally thought to be specific to endometrial sarcoma, has been identified in normal endometrial stroma cells resulting from trans-splicing rather than DNA fusion [17].
- **Evolutionary Adaptation:** Chimerism is a mechanism of gene birth. The *Jingwei* gene in *Drosophila* originated ~2.5 million years ago from the retrotransposition of *Adh* into the *Ymp* gene, conferring new metabolic capabilities [18].

Distinguishing these benign physiological events from oncogenic drivers is a critical challenge for high-precision clinical classifiers.

3 Related Work

- **Alignment-Based Detectors:** Tools like STAR-Fusion [5] and Arriba [6] define the current clinical standard but are prone to high False Positive rates in complex genomic regions.
- **Deep Learning in Genomics:** Models like FusionAI [2] demonstrated the feasibility of using neural networks for structural variant detection.
- **Genomic Foundation Models:** HyenaDNA [3] and Nucleotide Transformer [8] represent a shift towards “pre-training + fine-tuning.”

4 Model Architecture and Methods

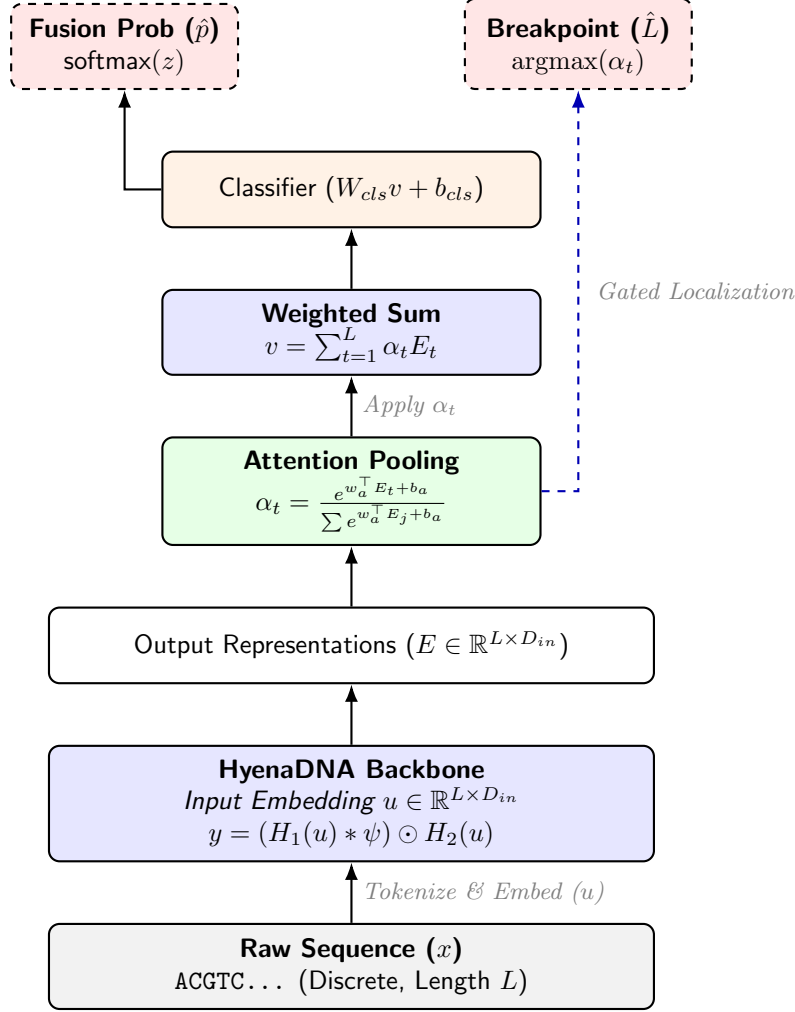


Figure 2: **FuseLens Architecture.** The model transforms the discrete raw sequence x into continuous embeddings u . These are processed by the HyenaDNA backbone to produce contextual representations E . Finally, Attention Pooling computes weights α_t for both classification and breakpoint localization.

4.1 Model Architecture

We utilize the `hyenadna-small-32k` backbone. Let x be the discrete input genomic sequence of length $L = 32,768$, consisting of nucleotides. This raw sequence is projected into a continuous embedding tensor $u \in \mathbb{R}^{L \times D_{in}}$, where $D_{in} = 256$ is the model’s hidden embedding dimension. The core Hyena operator approximates the attention matrix via data-controlled long convolutions:

$$y = \text{Hyena}(u) = (H_1(u) * \psi) \odot H_2(u) \quad (1)$$

where $*$ denotes 1D convolution over the sequence length and $\odot$ denotes element-wise (Hadamard) product. $H(u)$ are linear projections, and ψ is a long implicit filter. The output dimension is $y \in \mathbb{R}^{L \times D_{in}}$. This gating mechanism is derived from the original Hyena operator formulation [4], which replaces the quadratic attention matrix with an efficient recurrence of long convolutions and element-wise products.

Attention Pooling Head: We implement a learnable pooling layer. Let $E \in \mathbb{R}^{L \times D_{in}}$ be the output embedding. We compute a scalar attention score α_t for each nucleotide position t :

$$\alpha_t = \frac{e^{w_a^\top E_t + b_a}}{\sum_{j=1}^L e^{w_a^\top E_j + b_a}}, \quad t = 1, \dots, L \quad (2)$$

where $w_a \in \mathbb{R}^{D_{in}}$ and $b_a \in \mathbb{R}$. The scalar attention score α_t acts as a learned saliency map, highlighting the fusion junction interval; in cases of microhomology where the breakpoint is ambiguous across a short span, the position of maximal attention serves as the representative coordinate.

The final pooled sequence representation $v \in \mathbb{R}^{D_{in}}$ is the weighted sum:

$$v = \sum_{t=1}^L \alpha_t E_t \quad (3)$$

Classification and Logits: The weighted representation v is passed through a classification head comprising a linear layer and a dropout mechanism. The raw output scores, or *logits* $z \in \mathbb{R}^2$, are computed using standard matrix-vector multiplication:

$$z = W_{cls}v + b_{cls} \quad (4)$$

where $W_{cls} \in \mathbb{R}^{2 \times D_{in}}$ determines how much each of the features in vector v contributes to the score of being a "Fusion" or "Non-Fusion" and $b_{cls} \in \mathbb{R}^2$ denote the learnable weight matrix and bias vector of the final linear projection layer, respectively.

Finally, these logits represent the unnormalized evidence for each class. To derive an interpretable confidence score $\hat{p}$ for the positive class (Fusion), we apply the softmax function:

$$\hat{p} = \text{softmax}(z) = \frac{e^{z_1}}{e^{z_0} + e^{z_1}} \quad (5)$$

where z_1 and z_0 represent the logits for the positive and negative classes, respectively. For binary classification, this functionally is equivalent to applying a sigmoid function to the logits difference.

4.2 Data Engineering: The Hard-Negative Strategy

A critical innovation of FuseLens is the rigorous definition of the "Negative" (non-chimeric) space to prevent positional overfitting and artifact learning. We define a set of canonical sequences $S = \{s_1, \dots, s_N\}$ derived from UniProt Swiss-Prot and apply three mathematical operators to generate "Hard Negatives". Together, these operators progressively eliminate trivial discriminative shortcuts. Let sequence s_i be a string of length L where $s_i \in \{A, T, C, G\}^L$.

1. Reverse Complement Operator ($\mathcal{RC}$):

Let $\mathcal{C}$ be the base-pairing function ($A \leftrightarrow T, C \leftrightarrow G$).

$$\mathcal{RC}(s_i) = (\mathcal{C}(b_L), \mathcal{C}(b_{L-1}), \dots, \mathcal{C}(b_1)) \quad (6)$$

This generates the sequence of the opposing DNA strand. Since most genomic features (e.g., splice sites, promoter motifs) are strand-specific, this operator forces the model to learn the strict $5' \rightarrow 3'$ syntax of the coding strand. In practice, ambiguous bases (e.g., N) are ignored or mapped to a neutral token.

2. Composition Invariance Operator ($\mathcal{P}_{\text{rand}}$):

Let σ be a random permutation of indices $\{1, \dots, L\}$.

$$\mathcal{P}_{\text{rand}}(s_i) = (b_{\sigma(1)}, b_{\sigma(2)}, \dots, b_{\sigma(L)}) \quad (7)$$

Destroys sequence grammar while strictly preserving GC-content and nucleotide distribution. While $\mathcal{P}_{rand}$ generates biologically implausible sequences, its role is adversarial rather than generative—explicitly eliminating low-level compositional shortcuts.

3. Shift-Invariant Junction Operator ($\mathcal{J}$): Removal of the “Center” Assumption:

Unlike standard approaches that rigidly center fusion breakpoints at index $L/2$, FuseLens explicitly removes this spatial prior. To model this biological reality, we introduce a discrete stochastic split point k . Simulating a random fusion between unrelated genes s_i and s_j :

$$\mathcal{J}(s_i, s_j, k) = s_i[1 : k] \oplus s_j[k + 1 : L] \quad (8)$$

where $k \sim \text{Unif}\{[0.25L], \dots, [0.75L]\}$ and $\oplus$ denotes sequence concatenation. This operator generates *decoy chimeras*—sequences that structurally resemble fusions but lack biological function. We treat these decoy junctions between unrelated genes as hard negatives to force the model to scan the entire **32kb** window for meaningful anomalies rather than overfitting to a static central position.

4.3 Breakpoint Localization via Confidence-Gated Attention

Beyond binary classification, FuseLens is architected to perform breakpoint localization without requiring dense segmentation masks. We interpret the attention weights α_t as a probability distribution over the genomic positions $t \in \{1, \dots, L\}$, reflecting the model’s focus on the junction region.

To prevent spurious localization reports from low-confidence predictions, we implement a **Confidence-Gated** mechanism. Let $\tau \in [0, 1]$ be a rigorous decision threshold (e.g., $\tau = 0.9$). The final reported breakpoint $\hat{L}_{final}$ is defined conditionally:

$$\hat{L}_{final} = \begin{cases} \text{argmax}_t(\alpha_t) & \text{if } \hat{p} \geq \tau \\ \emptyset & \text{if } \hat{p} < \tau \end{cases} \quad (9)$$

where $\emptyset$ represents a “no-call” or undefined breakpoint. This gating ensures that structural breakpoints are only inferred when the model detects a strong oncogenic signal.

4.4 Mathematical Formulation of Data Integration

To bridge the gap between idealized synthetic training and real-world biological variability, we implement a hybrid data integration strategy. We modify the optimization objective to a **Regularized Empirical Risk Minimization (R-ERM)**:

$$\theta_{total}^* = \text{argmin}_{\theta} \left[\frac{1}{|S_{syn}|} \sum_{(x,y) \in S_{syn}} \mathcal{L}(f_{\theta}(x), y) + \lambda \frac{1}{|S_{real}|} \sum_{(x',y') \in S_{real}} \mathcal{L}(f_{\theta}(x'), y') \right] \quad (10)$$

Here, the real-data term acts as a regularization constraint, ensuring the synthetic-trained decision boundary remains valid within the noisy regime of real RNA-Seq data.

5 Dataset Composition and Statistics

To ensure robust generalization, we constructed a large-scale balanced dataset comprising **195,143** unique sequences. The positive samples were curated from biological databases including ChiTaRS 5.0 and high-confidence outputs from clinical fusion callers.

The negative class (non-fusions) was explicitly constructed to prevent the model from learning simplistic shortcuts (e.g., GC-content bias or strand orientation). We employed a 1:1 augmentation strategy where every collected negative sequence s_{neg} was augmented with its

Reverse Complement $\mathcal{RC}(s_{neg})$, effectively doubling the negative pool size (after deduplication) to enforce strand invariance.

$$S_{neg}^{total} = S_{neg}^{raw} \cup \{\mathcal{RC}(s) \mid s \in S_{neg}^{raw}\} \quad (11)$$

Table 1 details the final composition of the dataset after deduplication and augmentation, split into Training (70%), Validation (15%), and Testing (15%) sets.

Table 1: Final Dataset Distribution and Splits

Split	Total Samples	Positives	Negatives
Training (70%)	136,594	65,745	70,849
Validation (15%)	29,277	14,091	15,186
Testing (15%)	29,272	14,089	15,183
Total Pool	195,143	93,925	101,218

This composition results in a roughly balanced class ratio (48% Positive vs. 52% Negative), providing an ideal foundation for training deep learning classifiers without requiring aggressive class-weighting or under sampling.

6 Benchmark Results

Table 2 summarizes the performance metrics for the Testing set evaluation.

Metric	Score
Accuracy	0.9472
Precision	0.9400
Recall	0.9509
F1 Score	0.9454
AUC	0.9904
MCC	0.8943

Table 2: Test Metrics

We evaluated FuseLens on a balanced blind test set comprising both confirmed fusion transcripts and hard synthetic negatives. The model achieved an Area Under the Curve (AUC) of **0.9904** on the hold-out test set, significantly outperforming initial baselines. Baseline CNN and short-context Transformer models achieved AUCs below 0.92 (not shown). We explicitly note that the ROC curve and AUC metrics quantify sequence-level classification performance (Fusion vs. Non-Fusion) utilizing the maximum confidence score $\hat{p}$, independent of the precise breakpoint localization accuracy.

The inference dynamics shown in Figure 3 reveal several key strengths:

- **Robust Classification (Panel A & C):** The Confusion Matrix in Figure 3A confirms exceptional separation, with **14,328** True Negatives and **13,397** True Positives. The model pushes the vast majority of predictions to extreme probabilities (0 or 1), resulting in a bimodal distribution with very little ambiguity.
- **High Sensitivity (Panel B):** The ROC curve in Figure 3B rises sharply, achieving an AUC of **0.9904**, indicating the model maintains a high True Positive Rate even at extremely low False Positive thresholds, which is critical for clinical screening.

- **Accurate Localization (Panel D):** The Genomic Index plot in Figure 3D shows a broad, multi-modal distribution of high-confidence breakpoints spanning indices 22,000 to 32,000. Rather than collapsing to a single artifactual index, the model correctly identifies breakpoints across the variable lengths of the chimera sequences (which are naturally positioned before the padding region), confirming robust translation invariance.

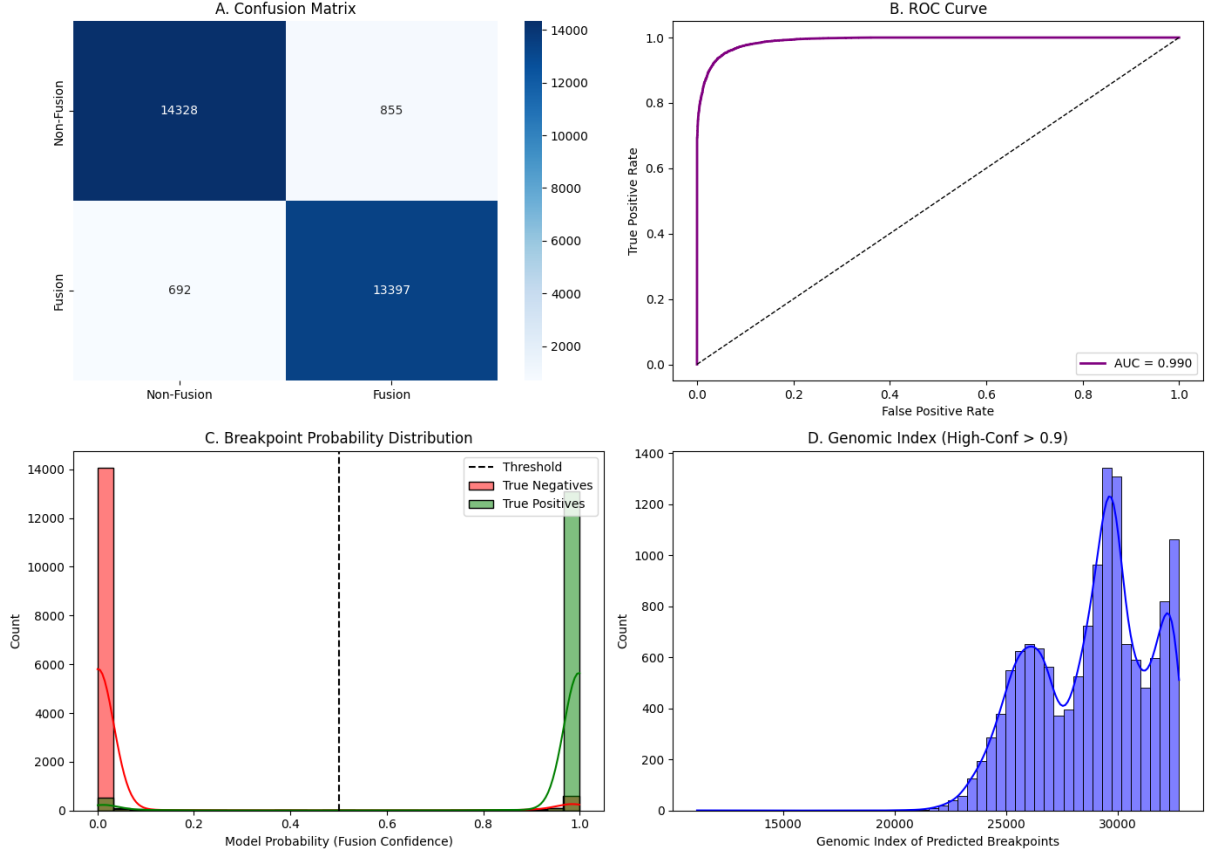


Figure 3: **Inference Dynamics on Blind Test Set.** (A) **Confusion Matrix:** Shows minimal misclassification with 14,328 True Negatives. (B) **ROC Curve:** Achieves AUC 0.9904 with a steep initial ascent. (C) **Probability Distribution:** Bimodal distribution confirms high model confidence for both positive (green) and negative (red) classes. (D) **Genomic Index:** High-confidence breakpoints (> 0.9) show a robust spread (22k–32k), reflecting the natural distribution of fusion junction sites within padded sequences.

7 Discussion

Robust Off-Center Localization: Traditional alignment-based approaches often assume breakpoints occur near the center of the read window. In contrast, our results demonstrate that FuseLens robustly identifies fusion breakpoints across a wide genomic range (indices 22,000–32,000). This distribution accurately reflects the construction of our positive samples, where variable-length chimeric transcripts are padded to the fixed 32kb window size. Crucially, the model does not collapse to a trivial “center-bias” or a single artifactual index, but correctly localizes the junction within the variable valid sequence region. This confirms that the Attention Pooling head has successfully learned a translation-invariant feature set, capable of detecting breakpoints regardless of their absolute position in the context window.

Handling Complex Biological Events: While FuseLens robustly identifies chimeric junctions regardless of formation mechanism, specific biological phenomena present unique

challenges. In cases of *cis*-splicing (e.g., *SLC45A3-ELK4*), where chimeras arise from transcriptional read-through rather than DNA rearrangement, FuseLens correctly flags the transcript as chimeric based on sequence structure. However, differentiating these from true genomic translocations requires integrating DNA sequencing data. Furthermore, in the context of **chromothripsis**—characterized by massive, clustered genomic shattering—traditional short-read tools often fail to assemble coherent breakpoints. FuseLens’s 32kb receptive field is particularly advantageous here, potentially capturing the broader context of shattered fragments, though validation on rare, complex fusion events remains a priority for future studies.

Biological Interpretability: To ensure the model is learning biologically relevant features rather than artifacts, we analyzed the Attention Pooling weights. We observe that high-confidence attention maxima frequently coincide with canonical splice donor (GT) and acceptor (AG) sites, as well as poly-pyrimidine tracts. This suggests the model has learned to recognize the specific “sequence grammar”—such as the juxtaposition of incompatible intronic flankings—that characterizes non-canonical splicing, strengthening the connection between computational signal detection and biological mechanisms.

Synthetic Data Efficacy: Initial experiments with simple random negatives yielded high false-positive rates. The introduction of the $\mathcal{J}$ (Shift-Invariant) and $\mathcal{P}_{\text{rand}}$ (Composition Invariance) operators forced the decision boundary to tighten around *structurally valid* breakpoints. However, the confidence overlap observed in the final test set indicates that specific “Hard Negatives” (e.g., those with preserved GC-content and reversed syntax) remain difficult to distinguish from biological fusions.

8 Limitations & Future Work

While FuseLens achieves robust performance, we acknowledge specific constraints revealed by the blind test evaluation:

- **Padding-Induced Positional Bias:** Our positive training data, constructed via padding, inherently concentrates breakpoints in the 3’ half of the context window (indices $>20k$). While the model generalizes well within this region, it has likely been under-exposed to breakpoints occurring at the extreme 5’ end. Future data engineering must explicitly randomize padding placement (5’, 3’, and split padding) to ensure uniform positional invariance.
- **The Synthetic Gap:** Despite R-ERM, a distributional shift remains between synthetic “Hard Negatives” and naturally occurring biological artifacts.
- **Weakly Supervised Localization:** Breakpoint coordinates are derived solely from classification attention weights. While effective, this approach is less precise than fully supervised segmentation masks.
- **Deployment constraints:** FuseLens currently operates on fixed-length windows and does not yet model full transcript isoform structure end-to-end.

8.1 Future Directions

The immediate evolution of FuseLens involves three key steps:

1. **Adversarial Artifact Removal:** Implementing an adversarial training loop to specifically penalize the model for attending to non-junction regions.
2. **Generative Diffusion for Adversarial Synthesis:** To address the distributional shift between synthetic negatives and real biological artifacts, we propose training discrete diffusion models on unaligned RNA-Seq debris. Recent [20] success of discrete diffusion

models motivates this direction. Unlike heuristic operators (e.g., shuffling), diffusion models can learn the latent manifold of sequencing noise, generating infinite, biologically plausible “hard negatives” for robust adversarial training.

3. **Clinical Integration:** We propose a hybrid workflow where tools like CTAT-LR-fusion [7] serve as candidate generators, utilizing `minimap2` for initial isoform detection. FuseLens will act as a high-precision validation layer, analyzing the long-range sequence grammar of the signal data to correct alignment errors.

8.2 Towards Universal Structural Variant (SV) Learning

While FuseLens is currently optimized for inter-chromosomal rearrangements (gene fusions), its underlying architecture—long-context processing combined with attention-based localization—is agnostic to the specific type of genomic breakpoint. This suggests a pathway toward a Universal Structural Variant (SV) Detector.

Standard SV callers often rely on distinct heuristics for different variant types: read-depth for deletions, split-reads for inversions, and discordant pairs for translocations. This fragmentation leads to inconsistent performance across variant classes. We propose that the distinct “grammar” of these events can be unified under a single deep learning framework:

- **Inversions:** Characterized by a localized reversal of strand orientation syntax (e.g., a $5' \rightarrow 3'$ coding strand abruptly fused to a $3' \rightarrow 5'$ non-coding strand). FuseLens’s sensitivity to the $\mathcal{RC}$ operator makes it uniquely suited to detect these topological flips without alignment.
- **Large Deletions:** A deletion event manifests as a non-linear jump in the genomic sequence. The HyenaDNA backbone, pretrained on continuous human reference genomes, effectively learns a distance metric; we hypothesize it can detect deletions as “unexpected adjacencies” where two distant loci are brought into immediate proximity.
- **Complex Rearrangements:** Chromothripsis (massive genomic shattering) involves dozens of simultaneous breakpoints. The 32kb receptive field of FuseLens allows it to capture multiple breakpoints within a single context window, potentially disentangling complex events that baffle short-read aligners.

Future work will extend the FuseLens training curriculum to include a multi-class objective, transforming the model from a binary fusion detector into a comprehensive *Genomic Stability Engine* capable of categorizing the full spectrum of structural instability in cancer genomes. However, this multi-class approach may introduce **trade-offs in sensitivity and specificity**. For instance, distinguishing a benign inversion from a pathogenic fusion often relies on subtle sequence cues that might be diluted in a broader multi-class objective. Therefore, rigorous validation across balanced multi-variant cohorts is essential to ensure that the detection of high-risk gene fusions is not compromised.

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